QUANT UX CASE STUDY · ARCADIA GROUP × QUBIT

What Analytics Couldn't Tell Us

Pairing exit intercepts with behavioral data at Topshop — and turning the answer into +5.8% conversion



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The decision at stake

Platform context. Qubit's Visitor Cloud unified analytics, segmentation, A/B testing and personalization for all seven Arcadia brands — experiments deployed as JS/CSS variations, measured against configurable statistical thresholds.

The fork in the road. After a header refresh, search engagement was underperforming and abandonment ticked up — and a sitewide search redesign was proposed on intuition. Scarce IT resources meant choosing: fund the redesign, or fund competing roadmap items. Nobody could say which, because nobody knew WHY search was failing.

My role. Analyst / optimization specialist: own the diagnosis, design the evidence, carry it to the prioritization table.

~10×

higher conversion for sessions that used search vs. sessions that didn't

Search wasn't a convenience feature — it was the single highest-intent behavior on the site. Friction there was directly suppressing revenue.

What the behavioral data could — and couldn't — say

DO · BEHAVIOR

Quantitative signals (Qubit analytics)

- Search adoption low relative to comparable retailers
- Sessions using search converted ~10× higher than sessions that never touched it
- Funnel analysis located drop-off; segment cuts ruled out device- or channel-specific causes

The structural limit

Analytics located the problem and priced it. It could not choose between three rival causes — and each implies a different, expensive fix.

Discoverability

Users don't see the search field

Fix: visual redesign

Usability

Users see it but can't use it well

Fix: interaction redesign

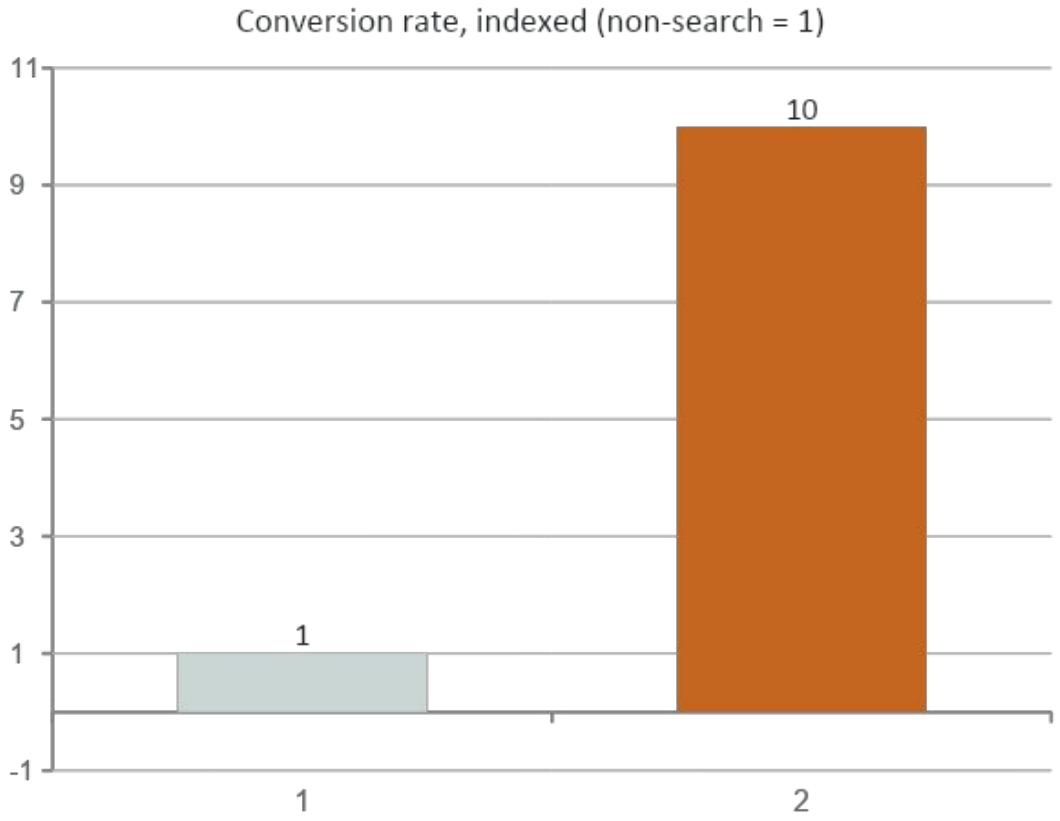
Result quality

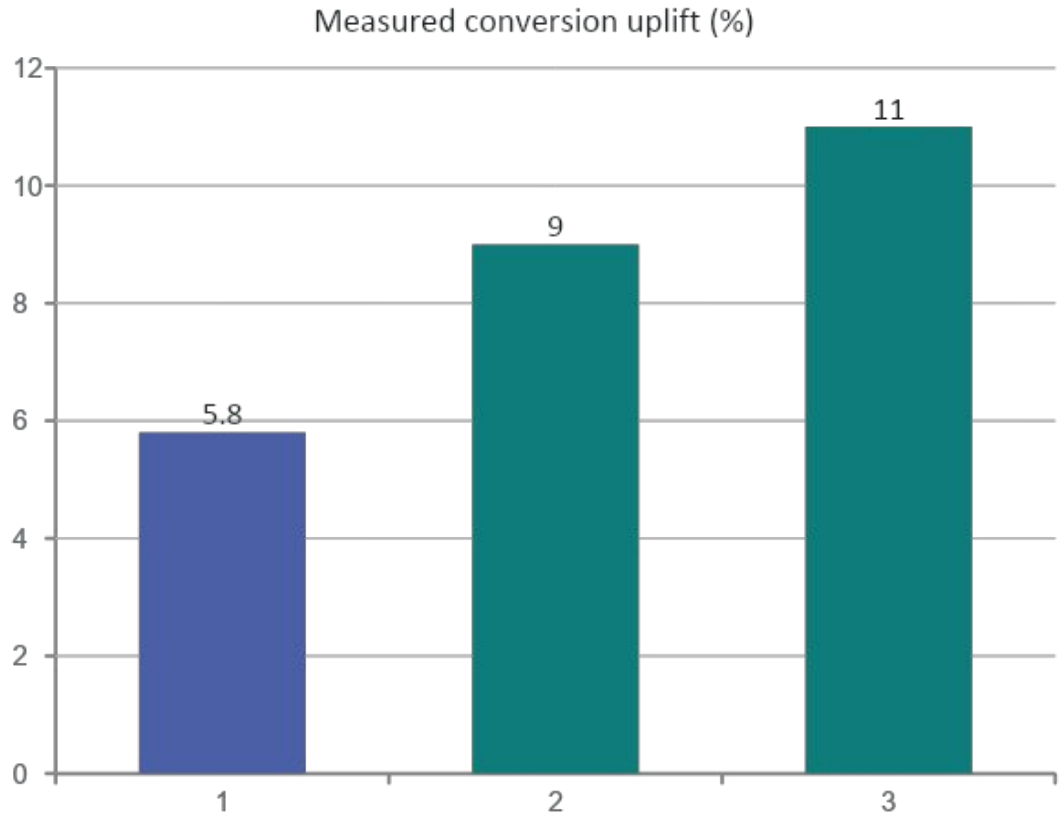
Search works; results disappoint

Fix: search backend

The analytics view

DO · BEHAVIOR





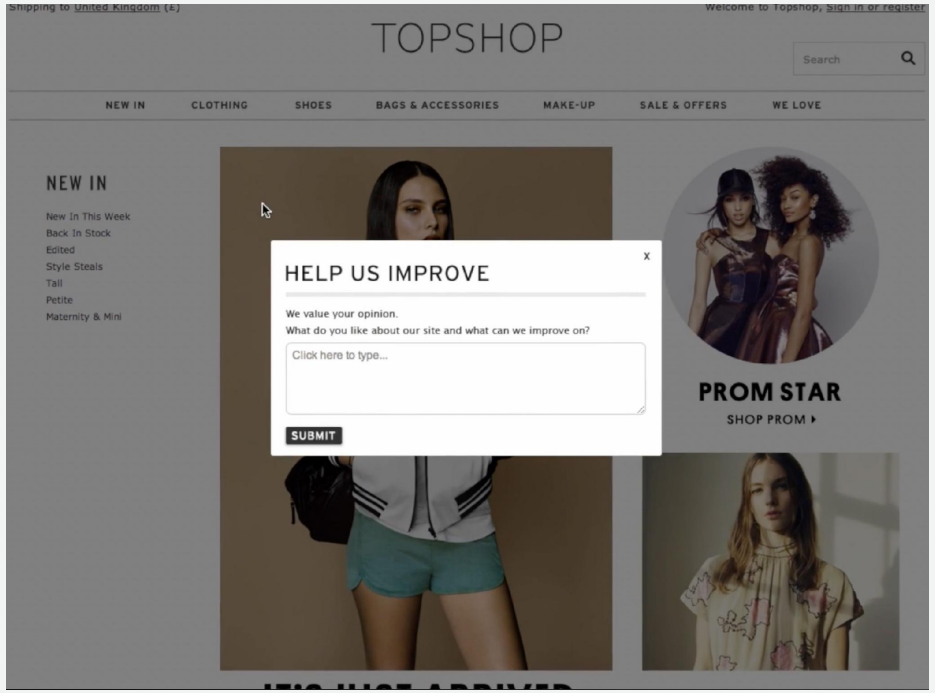
Indexed / reconstructed from reported metrics for presentation; program uplift measured as cumulative RPV vs. pre-period.

The intercept: capturing the why in-session

SAY · SURVEY

Instrument	Qubit Visitor Opinion, on-site intercept. Qualtrics today: Website / App Insights.
Trigger	Fired in-session at the point of friction and on exit intent. Qualtrics today: intercept action logic.
Format	Open-ended — the hypothesis space was unknown; closed options would only echo our guesses.
Targeting	Who, when, how often — frequency caps so heavy visitors aren't oversampled. Qualtrics today: embedded data + XM Directory rules.

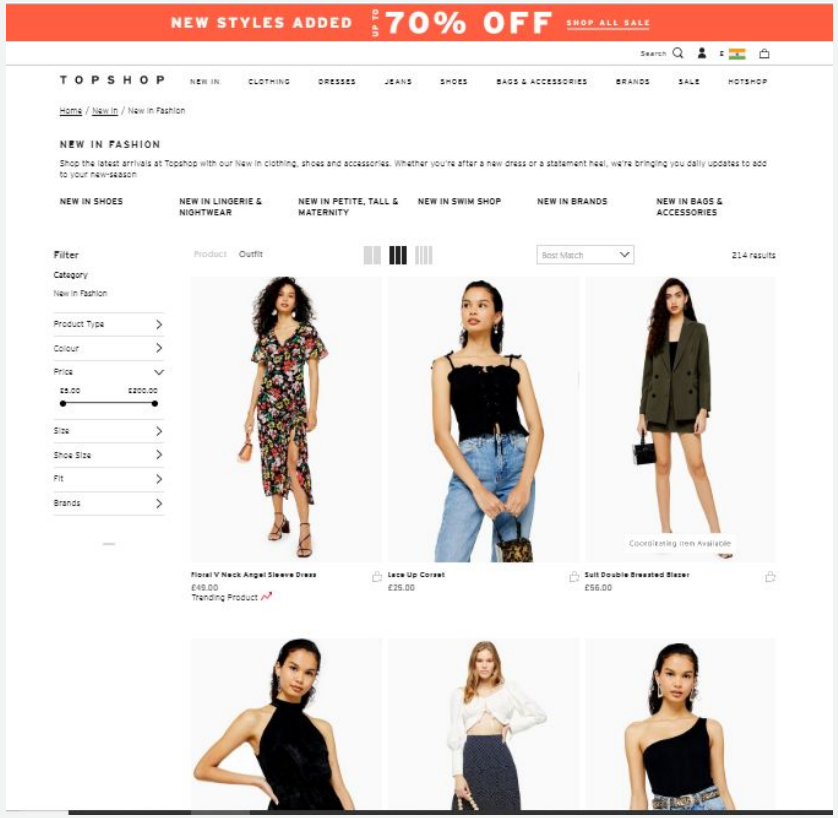
The recurring theme: users struggled to find and use the search bar — a low-contrast element with no affordance saying "type here." A causal hypothesis analytics could not produce.



The actual instrument, live on Topshop: an open-text Visitor Opinion intercept — in-session, unprompted, no checkboxes to bias the answer.

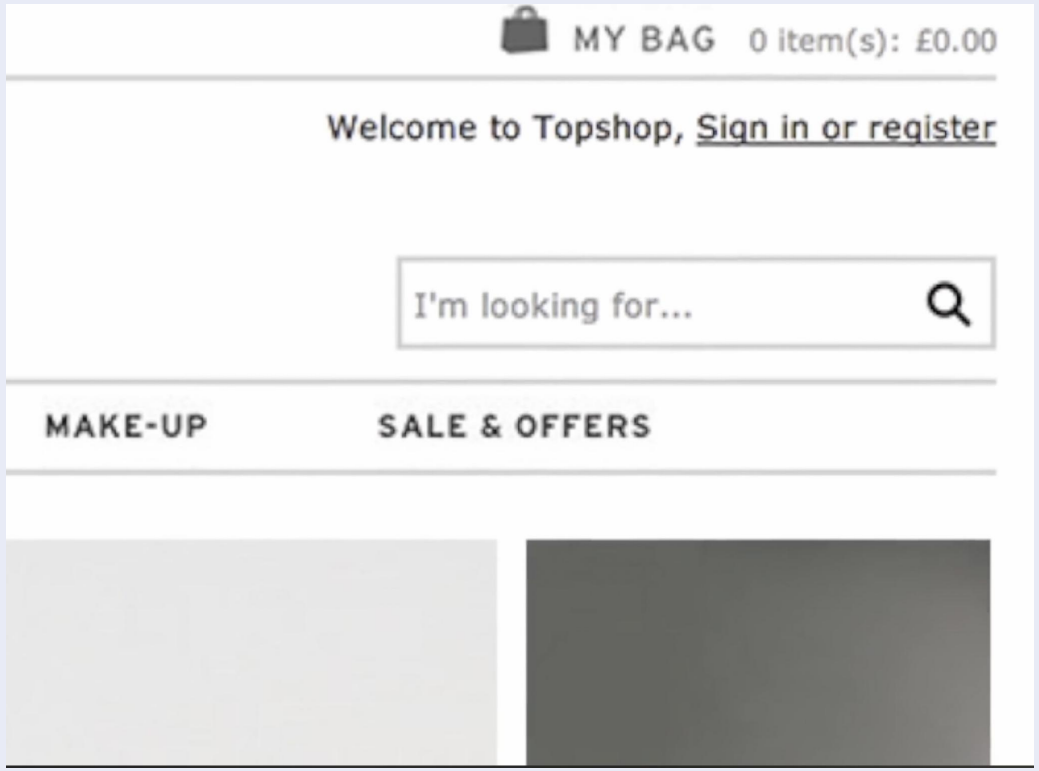
The surface in question

The header, as users met it



Search sits in the header competing with promo banners, nav, account, and bag — a text element among many

The shipped treatment (Variant D)



Triangulation: neither signal alone justified action

SAY — the diagnostic

"I can't find the search bar" — a mechanism. But alone, it's an anecdote that loses roadmap arguments to flashier ideas.

+

DO — the business case

Search users convert ~10× — a prize. But alone, it's a problem with no actionable fix attached.

A sized, mechanism-specific, testable hypothesis

"Search visibility and affordance suppress adoption; fixing them will lift conversion — and the lift is worth engineering time, because the affected behavior converts at 10×."

Design the inference, then run the test

TEST · CAUSAL

Two mechanisms from feedback → 2×2 factorial structure (+ control)

A — Copy change only

Invitation hypothesis, isolated

B — Border only

Visibility hypothesis, isolated

C — Border + expanded copy

Combined effect

D — Border + concise copy

Combined, minimal ★ winner

vs. CONTROL — existing design, evenly split traffic

Pre-committed decision rules

- Even randomized split across all five cells — assignment by visitor, consistent within session
- Primary metric: conversion — chosen because the 10× stakes made it directly revenue-bearing
- Run to statistical significance BEFORE any rollout conversation — evidence, not opinion, goes to IT prioritization
- Factorial structure means the winner is attributable to a mechanism — not to novelty

Today I'd add: pre-registered duration (no peeking), and explicit guardrail metrics — abandonment, complaint rate via a light intercept.

Statistical validation: signal, not noise

TEST · CAUSAL

Variant	Mechanism tested	Conversions / visitors	CR	Rel. uplift	Significant?
Control	Existing UI — no border, default copy	13,020 / 620,000	2.10%	baseline	—
A — Copy only	“Invitation”, isolated	13,082 / 620,000	2.11%	+0.5%	No — n.s. ($p > .05$)
B — Border only	“Visibility”, isolated	13,640 / 620,000	2.20%	+4.8%	Yes ($p < .05$)
C — Border + long copy	Combined, expanded	13,516 / 620,000	2.18%	+3.8%	Yes ($p < .05$)
D — Border + concise copy	**Combined, minimal**	**13,776 / 620,000**	**2.22%**	**+5.8%**	**Yes — winner**

Pre-committed inference: two-proportion Z-test per cell against control, 95% threshold fixed before launch — no peeking, no early stopping.

The attribution readout: A flatlined against control; every bordered cell jumped. Visibility carried the effect — invitation never mattered.

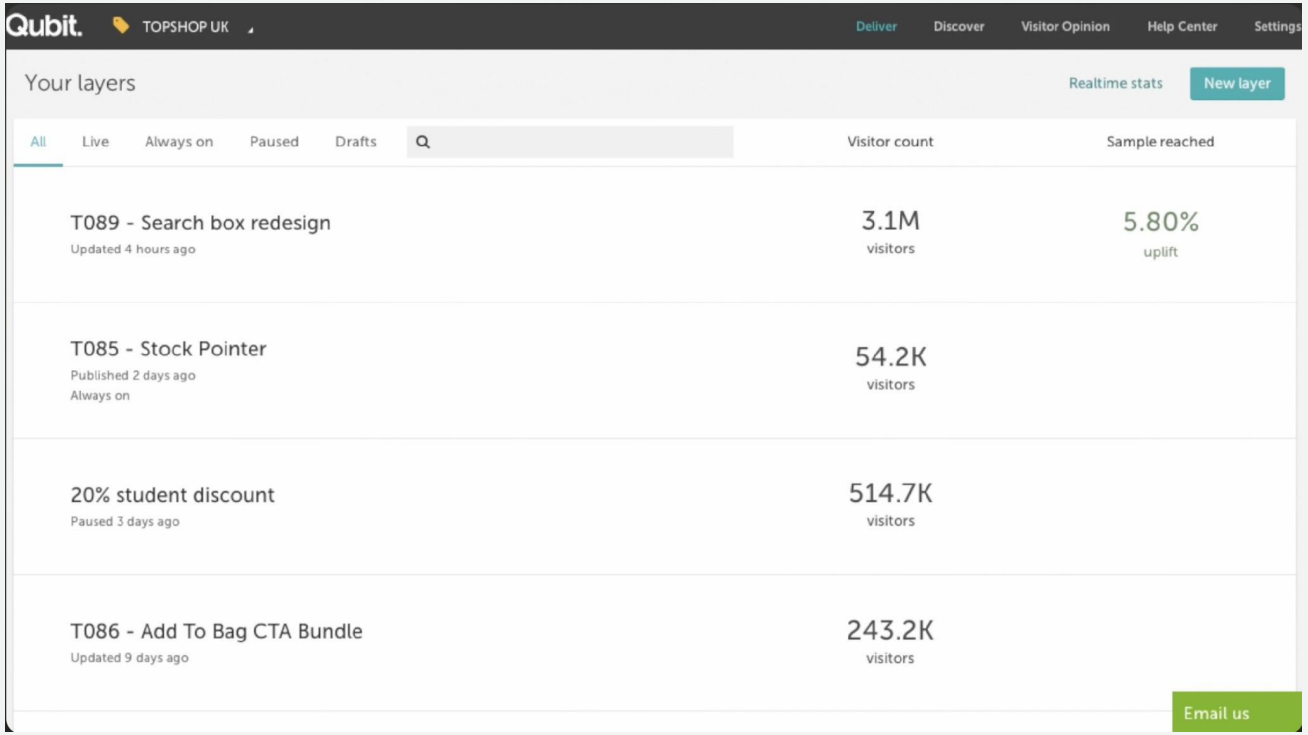
Representative figures, reconstructed to be consistent with the reported +5.8% uplift and the ~3.1M-visitor test scale shown on the platform dashboard; raw exports remain internal to Qubit / Arcadia. $Z = (CR_V - CR_C) / SE$, $SE = \sqrt{\bar{p}(1-\bar{p})(1/n_C + 1/n_V)}$.

Results: the platform receipt

+5.8%

conversion uplift on 3.1M visitors — Variant D, shipped sitewide

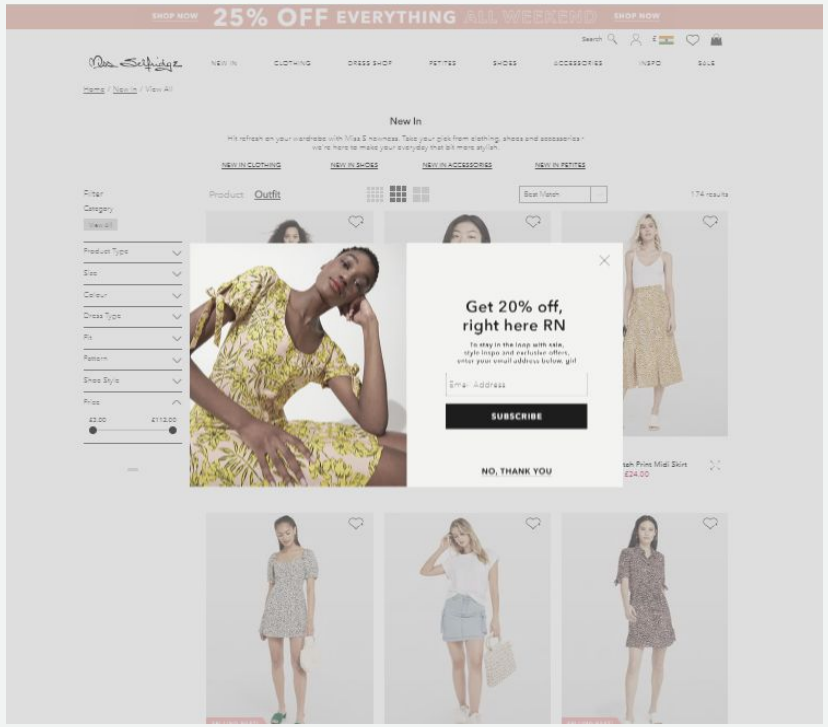
- Border carried the effect → visibility was the binding constraint
- Copy contributed nothing → the equally "obvious" rival hypothesis lost
- Generalisable design constraint: every low-affordance element became a suspect



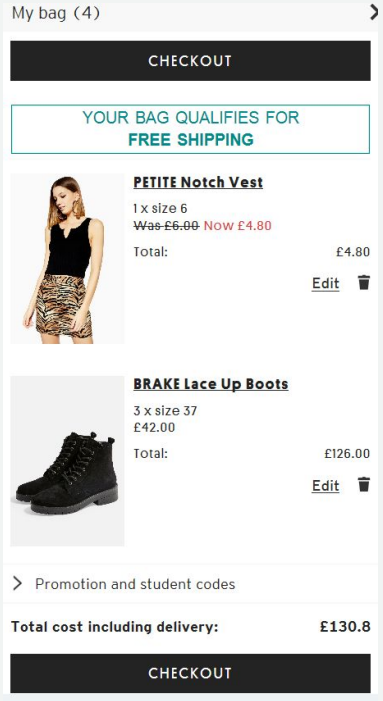
Qubit experimentation dashboard — T089 Search box redesign: 3.1M visitors, 5.80% uplift. T085 Stock Pointer and T086 Add-to-Bag CTA below it are the same loop, running on.

The loop compounds: the same intercept → hypothesis → test cycle across product-page micro-changes added 9–11% cumulative — tracked as cumulative RPV, because sequential results don't naively sum.

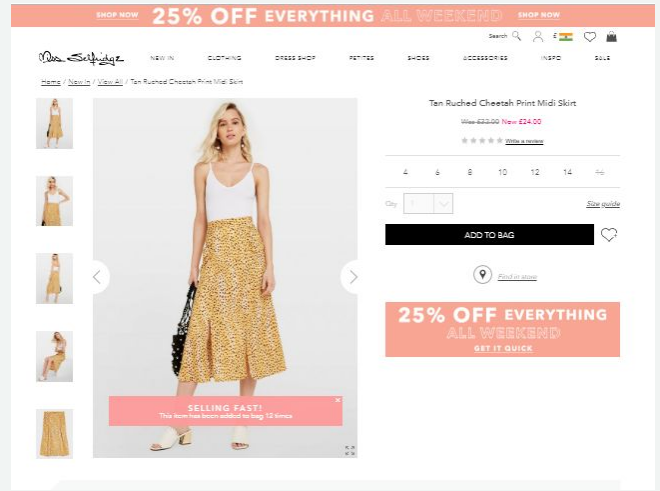
The compounding loop, visually



Product page micro-changes: “selling fast” urgency + “only 1 left” stock signal — each one tested, not assumed



Recently-viewed slider — recovery of intent



Signup pop-up — targeted by visitor state, capped by frequency rules; the same targeting-logic muscle as the intercepts

Limitations — and what I'd do differently today

Intercept sampling skews engaged

Exit intercepts oversample invested users; the silently-bouncing majority is underheard.

Today: frequency caps + quota targeting in Qualtrics; weight responses by session-frequency from behavioral data.

Manual open-end coding

Theming hundreds of verbatims by hand is slow and subjectivity-prone.

Today: NLP-assisted clustering for scale, human review for validity — the exact hybrid I build at ABIM.

Single primary metric

Conversion-only optimization can hide experience costs elsewhere.

Today: explicit guardrails — abandonment, rage-clicks, complaint-rate intercept — defined before launch.

Does the discipline transfer? B2B, small-N, same rigor

John Deere — longitudinal usability study, enterprise migration

60 → 85.6

SUS across the study window — vs. industry average 68, 'excellent' threshold 80

N=4 + N=40

power users in longitudinal depth, triangulated with an org-wide survey

Novelty handled

familiarization window before measuring — day-one scores measure the learning curve, not the product

Same loop, your stack

Qubit behavioral analytics	→	Glassbox: funnels, struggle detection, session replay
Qubit Visitor Opinion intercepts	→	Qualtrics: intercepts, embedded data, XM Directory, dashboards
Qubit experimentation	→	Your testing platform + my analysis layer (Python/R)

The loop I'd bring to this role

GLASSBOX	behavioral data finds the where — funnels, struggle, replay
QUALTRICS	intercepts find the why — triggers, embedded data, directories
EXPERIMENTS	prove the fix — designed inference, practical significance
DASHBOARDS	keep it honest — tracked metrics, compounding wins

Numbers tell you the scale. Feedback tells you the cause. Experiments tell you the truth.